Brain activity rides the connectome's harmonics, and psychedelics bend which ones: a parcellated reanalysis of LSD and psilocybin fMRI

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Connectome Harmonics open-source project · github.com/fractastical/connectome_harmonics

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Abstract

Connectome harmonics — the eigenmodes of the graph Laplacian of the human structural connectome — provide a frequency-ordered basis for brain activity. We combine a group-average Human Connectome Project (HCP) Schaefer-400 structural connectome with two independent within-subject psychedelic fMRI datasets (LSD: OpenNeuro `ds003059`, $n = 12$; psilocybin: OpenNeuro `ds006072`, $n = 7$) and ask four questions. (1) *Does LSD reorganize the harmonic spectrum?* At parcellated scale low-frequency harmonics lose power under LSD (low-band LSD/placebo power ratio ≈ 0.77), reproducing the direction of Atasoy et al. (2017). (2) *Is there a geometric latent space?* Using the lowest harmonics as coordinate axes, the seven Yeo functional networks separate into a continuous, unsupervised layout that emerges from connectivity alone. (3) *Shape versus wiring — and does a psychedelic move the winner?* Reconstructing the BOLD signal from connectome, cortical-geometry (Laplace–Beltrami), and exponential-distance-rule (EDR) bases, all three are close at 400-region scale, but under LSD the geometric basis reliably gains on the connectome in a within-subject paired test (ΔAUC $= +0.0058$, 95% CI $[+0.0030, +0.0085]$, paired $t(11) = 4.60$, $p = 0.0008$; exact sign-flip permutation $p = 0.001$; Cohen's $d_z = 1.33$; 12/12 subjects), and this shift survives global-signal regression and frame censoring ($p = 0.002$, permutation $p = 0.003$). We then *replicate* this effect in an independent psilocybin-versus-active-placebo cohort: the same direction and a large effect (ΔAUC $= +0.010$, $d_z = 0.87$), significant by exact permutation and Wilcoxon tests ($p = 0.031$), borderline parametrically (t-test $p = 0.061$) and attenuating below threshold under aggressive denoising. At vertex resolution (fsLR-32k), using Pang's *empirical* high-resolution group-average connectome, the geometric basis reconstructs cortical activity better than the real connectome in both states, confirming the geometric advantage off the parcellation; the drug-induced shift is directionally consistent but underpowered ($n = 7$) and not significant at vertex scale, so the shift itself remains a parcellated-level result. (4) *Does the harmonic spectrum become more consonant/symmetric, as the Symmetry Theory of Valence would predict for a typically positively-valenced state?* Scoring the sensory dissonance, spectral entropy, and centroid of the connectome-harmonic power spectrum per subject and state, we find the *opposite*: under LSD the spectrum becomes significantly more dissonant ($p = 0.001$, $d_z = 1.24$, surviving denoising), broader, and higher-frequency, with psilocybin trending the same way — the "entropic brain" direction. We interpret these results as evidence that psychedelics shift cortical activity toward smoother, shape-aligned spatial patterns and away from specific wiring, even as the overall harmonic repertoire broadens and de-tunes, while emphasizing the coarse, parcellated resolution and modest sample sizes. All code, data-fetching scripts, and an interactive web demonstration are openly available.

Keywords: connectome harmonics; graph Laplacian eigenmodes; psychedelics; LSD; psilocybin; cortical geometry; functional connectivity; fMRI.

1 Introduction

The brain's structural wiring, the *connectome*, can be represented as a weighted graph whose nodes are brain regions and whose edges are anatomical connections. The eigenvectors of the graph Laplacian of this network form a set of *connectome harmonics*: spatial standing-wave patterns ordered from smooth

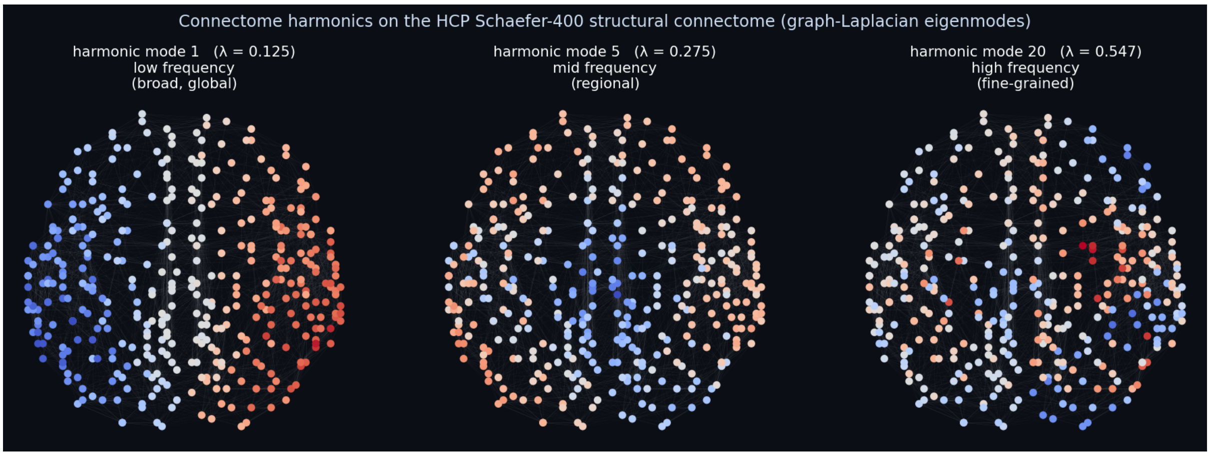

Figure 1: Connectome harmonics on the brain. Three graph-Laplacian eigenmodes of the HCP Schaefer-400 structural connectome, rendered on the cortical layout (red/blue = opposite sign of the standing-wave pattern; this is the basis the interactive demo animates). Low modes are broad and global, higher modes increasingly fine-grained; each mode's eigenvalue λ gives its spatial frequency. Brain activity is reconstructed as weighted sums of such modes.

and global (low spatial frequency) to fine-grained (high spatial frequency), analogous to the vibrational modes of a drum [1] (Figure 1). Resting-state and task fMRI can be decomposed into this basis, and Atasoy et al. [2] reported that lysergic acid diethylamide (LSD) reorganizes the harmonic repertoire, expanding the range of active modes.

A parallel line of work [4] has argued that the *geometry* of the cortical surface — captured by the eigenmodes of the Laplace–Beltrami operator (LBO) on the cortical mesh — predicts functional activity at least as well as connectivity-derived modes, prompting an ongoing debate about whether shape or wiring is the more fundamental constraint [5].

Here we unify these threads in a single, fully reproducible, parcellated pipeline. Working at the Schaefer-400 parcellation, we (i) measure the LSD-induced change in the connectome-harmonic power spectrum, (ii) test whether the low-dimensional harmonic embedding recovers known functional networks without supervision, and (iii) directly compare connectome, geometric, and distance-based bases for their ability to reconstruct the BOLD signal, asking whether the relative advantage of geometry over wiring *shifts* under a psychedelic. Crucially, we treat the within-subject design as the source of statistical power: every subject provides both drug and control states, so we test each subject's own drug-minus-control difference. We then ask the strongest question available to us — whether the LSD effect *replicates* in an independent psilocybin dataset acquired on a different scanner with a different preprocessing pipeline and an active (stimulant) placebo.

2 Data and Methods

2.1 Datasets

Structural connectome. A group-average HCP Young-Adult structural connectivity matrix, parcellated to the Schaefer-400 atlas (400 cortical regions, 7-network Yeo assignment), provides the weighted graph W.

LSD fMRI (`ds003059`). Resting-state fMRI from the Carhart-Harris LSD study [3], OpenNeuro `ds003059`, $n = 12$ within-subject (each subject scanned under LSD and placebo). Volumetric data were parcellated to Schaefer-400.

Psilocybin fMRI (`ds006072`). A precision-imaging psilocybin trial [6], OpenNeuro `ds006072`, $n = 7$ within-subject, comparing psilocybin (25 mg) against an *active* placebo (methylphenidate, 40 mg). We used the provided fsLR-32k surface CIFTI dense time series (bandpassed, no global-signal regression), parcellated to Schaefer-400 by averaging grayordinates within each parcel. This is a methodologically cleaner, surface-based path than the volumetric LSD pipeline, and an entirely independent cohort, scanner, and preprocessing stream — the key ingredients of a genuine replication.

Cortical geometry. Geometric eigenmodes were computed as LBO eigenmodes of the fsLR-32k midthickness surface (NSBLab/BrainEigenmodes [4]) using LaPy, then parcellated to Schaefer-400.

2.2 Bases

We compare three orthonormal bases on the 400 parcels:

- **Connectome harmonics**: eigenvectors of the symmetric normalized graph Laplacian

$$L = I - D^{-1/2}\, W\, D^{-1/2}, \tag{1}$$

  where D is the degree matrix of W. Eigenvalues order the harmonics by spatial frequency.

- **Geometric eigenmodes**: parcellated LBO eigenmodes of the cortical surface (shape, no wiring).

- **Exponential distance rule (EDR)**: eigenmodes of a surrogate connectome whose edge weights decay as $\exp(-d/\lambda)$ with inter-region distance d and length constant $\lambda = 20\,\text{mm}$ — geometry expressed as a network, with no real wiring.

2.3 Reconstruction-accuracy comparison

For each basis we reconstruct every parcellated BOLD frame from its first N modes and score the reconstruction by its spatial correlation with the original, averaged over frames, for N on a logarithmic grid $(1 \ldots 200)$. The area under this accuracy-versus-N curve (AUC) summarizes how efficiently a basis recreates the data: a more natural basis reaches high accuracy with fewer modes. For each subject and state we compute the *basis advantage* AUC(geometric) − AUC(connectome).

Two distinct contrasts (and why EDR is not wiring). It is essential to keep two comparisons separate. The *geometry-vs-wiring* contrast, AUC(geometric) − AUC(connectome), pits cortical *shape* against the *real structural connectome* — including its long-range tracts — and is the headline of this paper. The *geometry-vs-distance* contrast, AUC(geometric) − AUC(EDR), pits shape against a *distance-decay surrogate*; because EDR encodes only "connections fall off with distance" it is itself a geometric object, *not* a model of wiring. A shift in the first contrast is a statement about shape versus wiring; a shift in the second is a statement about two flavours of geometry and must not be read as a wiring result.

2.4 Harmonic consonance metric (Experiment 4)

The Symmetry Theory of Valence (STV) [7] posits that the pleasantness of an experience tracks the *symmetry* of its neural representation, and has been operationalized on connectome harmonics via musical-style *consonance*. Every metric below is a one-number summary of a single object, the *harmonic power spectrum* of a scan, which we build as follows.

Harmonic power spectrum. Let $Y \in \mathbb{R}^{400\times T}$ be the per-region z-scored BOLD and $V = [\,v_1, \ldots, v_K\,]$ the orthonormal connectome harmonics (eigenvectors of the graph Laplacian, eigenvalues $\lambda_1 \le \cdots \le \lambda_K$, $K = 200$). After removing each frame's spatial mean ($\tilde{Y}_{:,t} = Y_{:,t} - \bar{Y}_{:,t}$), we project onto the harmonics, $C = V^\top \tilde{Y}$, so $C_{k,t}$ is the instantaneous amplitude of mode k. The *relative power* of mode k is its time-averaged squared amplitude, normalized to a distribution:

$$p_k \;=\; \frac{\langle C_{k,t}^2\rangle_t}{\sum_j \langle C_{j,t}^2\rangle_t}, \qquad \sum_k p_k = 1. \tag{2}$$

Each mode also carries a spatial frequency $\omega_k = \sqrt{\lambda_k}$ (the vibrating-membrane analogy: low modes are smooth/global, high modes fine-grained). The pair $\{\omega_k, p_k\}$ is the spectrum; the three metrics are different summaries of it.

Spectral entropy and centroid. The normalized Shannon entropy

$$H \;=\; -\frac{1}{\ln K}\sum_k p_k \ln p_k \;\in [0,1] \tag{3}$$

measures how *spread out* the power is ($H \to 0$: concentrated in a few modes; $H \to 1$: a broad, flat repertoire). The spectral centroid $\bar{\omega} = \sum_k p_k\,\omega_k$ is the power-weighted mean spatial frequency (higher = energy in finer modes). We also report the ratio of power in the lowest to the highest third of modes.

Sethares consonance. Unlike H and $\bar{\omega}$, consonance depends on the *relationships between* modes. Treating the spectrum as a chord — each mode a tone of frequency f_k and amplitude $a_k = \sqrt{p_k}$ — we sum the Plomp–Levelt/Sethares [8] sensory dissonance over all pairs:

$$D \;=\; \sum_{i<j} \min(a_i, a_j)\Big(e^{-A_1 s_{ij}\,|f_i - f_j|} - e^{-A_2 s_{ij}\,|f_i - f_j|}\Big), \qquad s_{ij} = \frac{D^\star}{S_1 \min(f_i, f_j) + S_2}, \tag{4}$$

with Sethares' constants $A_1 = 3.51$, $A_2 = 5.75$, $D^\star = 0.24$, $S_1 = 0.0207$, $S_2 = 18.96$. The two-exponential kernel is zero for identical tones, peaks at a small separation (roughly one critical band), and decays for well-separated tones: power landing in *neighbouring* modes is penalized as "beating." Lower D = more consonant; we also report $1/(1+D)$. Because the Sethares constants are calibrated in Hz, we map ω_k proportionally into the audible range ($f_k = 220\,\omega_k/\min_k \omega_k$, lowest mode $\to$ 220 Hz); this preserves all frequency *ratios* and is applied identically to every subject and state, so it cannot bias the drug-vs-control contrast (we treat the absolute scaling as a modelling choice and verify robustness to it).

Each metric is submitted to the same within-subject paired test, exact sign-flip permutation null, and GSR robustness rerun as the basis advantage. We also recompute all metrics on the geometric harmonics as a substrate check.

2.5 Statistics

The within-subject *shift* is the per-subject change in basis advantage between drug and control (LSD−placebo, or psilocybin−methylphenidate). We report the mean shift with its 95% confidence interval, a paired t-test, the nonparametric Wilcoxon signed-rank test, Cohen's d_z effect size, and an *exact* within-subject sign-flip permutation test (all 2^n label assignments: 4096 for $n = 12$, 128 for $n = 7$). At small n the exact permutation and Wilcoxon tests are the appropriate inferential tools; we note that the smallest attainable two-sided permutation p-value is $2/2^n$ (≈ 0.0005 at $n = 12$, ≈ 0.016 at $n = 7$), so the psilocybin design is intrinsically power-limited.

2.6 Robustness / nuisance regression

Because the raw LSD dataset lacks fmriprep motion parameters, we apply the strongest nuisance model computable from the parcel time series alone: linear+quadratic detrending, global-signal regression (GSR, plus its temporal derivative), and DVARS-based Tukey-fence frame censoring. We then rerun the identical paired and permutation tests on the denoised data. GSR removes the dominant motion/arousal confound for drug-state contrasts, so survival of the effect after GSR is our honest bottom line.

2.7 Reproducibility

All analyses are scripted in Python (`numpy`, `scipy`, `nibabel`, `nilearn`, `lapy`) with pinned dependency versions. Data are fetched programmatically from their original sources. Code, precomputed results, and an interactive browser demonstration are available at https://github.com/fractastical/connectome_harmonics (live demo: https://fractastical.github.io/connectome_harmonics/).

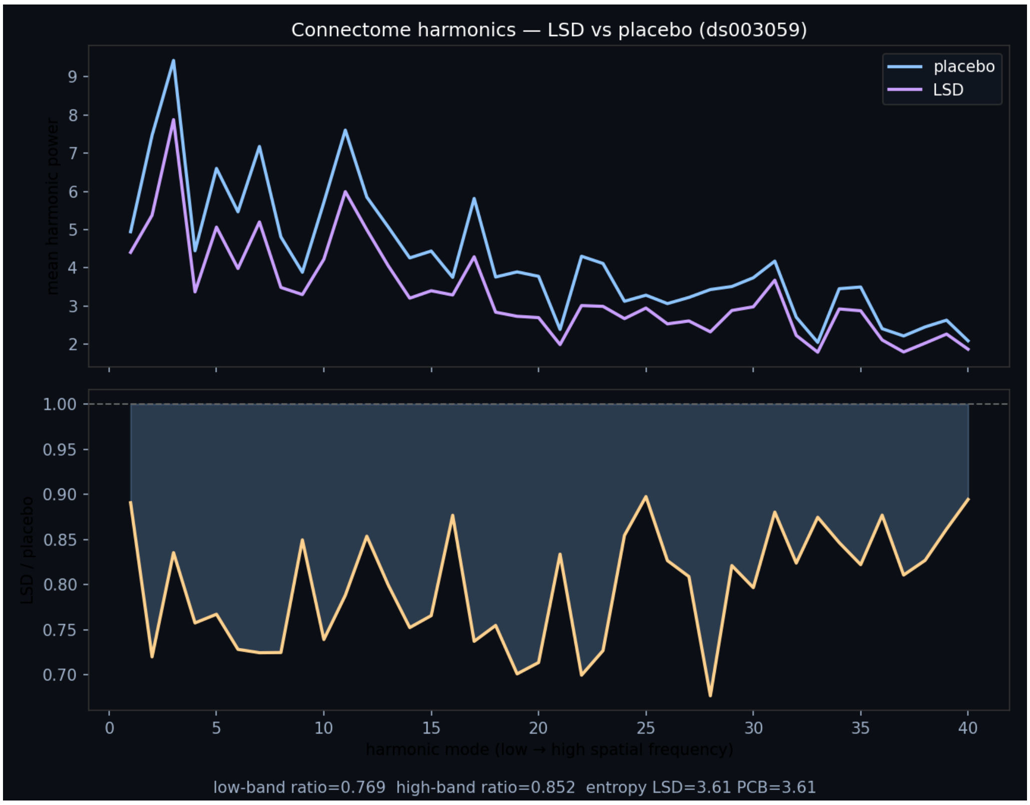

Figure 2: Experiment 1. Group-mean connectome-harmonic power versus spatial frequency (mode index), LSD versus placebo ($n = 12$, ds003059), with the per-mode LSD/placebo power ratio below. Low-frequency modes are suppressed under LSD (low-band ratio 0.77).

3 Results

3.1 Experiment 1: LSD reorganizes the harmonic spectrum (in direction)

Projecting ds003059 BOLD onto the HCP connectome harmonics, low-frequency harmonics carry less power under LSD: the group mean low-band LSD/placebo power ratio is 0.77 and the high-band ratio is 0.85 (Figure 2), reproducing the *direction* reported by Atasoy et al. [2]. Repertoire entropy is essentially unchanged at this scale (LSD 3.614 vs. placebo 3.607). The high-frequency power *increase* emphasized in the original vertex-level work does not clearly separate at the coarse parcellated resolution used here.

3.2 Experiment 2: an unsupervised geometric latent space

Using the lowest non-trivial harmonics as coordinate axes (a Laplacian eigenmap), all 400 regions embed into a low-dimensional space in which the seven Yeo functional networks separate into a continuous layout, ordered along the first harmonic axis as Default-mode → Dorsal-attention → Somatomotor → Salience/Ventral-attention → Limbic → Frontoparietal-control → Visual. No functional labels were provided to the embedding: the organization emerges from structural connectivity alone, consistent with the existence of a smooth functional gradient encoded geometrically in the connectome.

3.3 Experiment 3: shape versus wiring, and the LSD shift

At 400-region resolution all three bases reconstruct the BOLD signal comparably (Figure 3, left/middle), with the distance-only EDR basis achieving the best AUC in both states — geometry does at least as well as wiring (Table 1). The novel result is the *shift*: the per-subject geometric−connectome advantage is negative under placebo (-0.0021) but positive under LSD ($+0.0037$), a within-subject increase of $\Delta\text{AUC} = +0.0058$ (95% CI $[+0.0030, +0.0085]$; paired $t(11) = 4.60$, $p = 0.0008$; Wilcoxon $p = 0.001$; exact permutation $p = 0.001$; $d_z = 1.33$, large), with geometry gaining in all 12/12 subjects (Figure 3,

Table 1: Reconstruction AUC by basis and state. Higher is better. $\Delta_{\text{geo-conn}}$ is the geometric−connectome advantage.

Dataset	State	Connectome	Geometric	EDR
LSD (ds003059, n=12)	Placebo	0.7668	0.7647	**0.7817**
	LSD	0.7515	0.7552	**0.7730**
Psilocybin (ds006072, n=7)	Methylphenidate	0.7149	0.7051	**0.7120**
	Psilocybin	0.6914	0.6928	**0.7002**

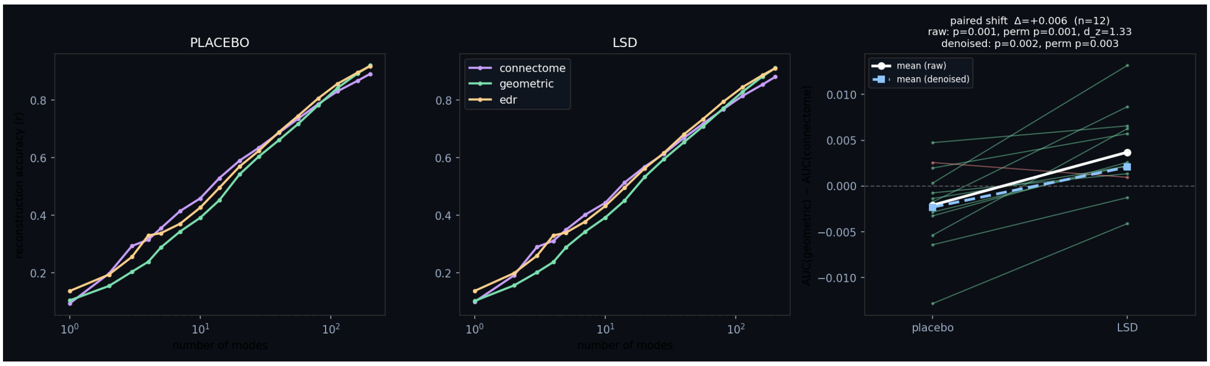

Figure 3: Experiment 3 (LSD). Reconstruction accuracy versus number of modes for each basis under placebo (left) and LSD (middle), and the per-subject paired shift of the geometric−connectome advantage (right). Nearly every subject moves toward geometry under LSD; the white/blue lines show the raw and nuisance-regressed group means.

right). After detrending, GSR, and DVARS censoring (98% of frames retained) the effect attenuates by roughly a quarter but remains significant ($\Delta\text{AUC} = +0.0044$, paired $p = 0.002$, permutation $p = 0.003$, $d_z = 1.19$): it is not merely a motion or global-signal artifact.

3.4 Replication: the shift recurs under psilocybin

We reran the identical paired analysis on the independent psilocybin cohort (ds006072; psilocybin vs. active-placebo methylphenidate; $n = 7$). The effect replicates *in direction* with a large effect size: the geometric−connectome advantage shifts toward geometry by $\Delta\text{AUC} = +0.0103$ (95% CI $[-0.0006, +0.0213]$; $d_z = 0.87$). It is significant by the appropriate nonparametric tests — exact sign-flip permutation $p = 0.031$ and Wilcoxon $p = 0.031$ — and borderline by the parametric paired t-test ($t(6) = 2.30$, $p = 0.061$). Under aggressive GSR the effect attenuates to $\Delta\text{AUC} = +0.0061$ ($d_z = 0.63$) and slips just below significance (permutation $p = 0.063$), mirroring the partial attenuation seen for LSD (Figure 4, Table 2). With $n = 7$ against an active stimulant placebo, the design is intrinsically underpowered; the converging *direction and effect size across two independent psychedelics* is the substantive result, not any single p-value.

3.5 Vertex-resolution robustness check

The principal limitation of the analyses above is their 400-region parcellation, the regime in which the geometric advantage is weakest. We therefore repeated the reconstruction comparison on the psilocybin cohort at *vertex* resolution, in fsaverage5 space (10,242 vertices, left hemisphere), the resolution at which Pang et al. [4] publish eigenbases. The fsLR-32k BOLD was resampled to fsaverage5 with a pure-Python area-average resampler built from registration-fusion spheres (validated against native Laplace–Beltrami modes, $|r| = 0.86$–0.97 for the lowest modes). Two findings emerge (Figure 5).

First, the **geometric basis reconstructs vertex BOLD efficiently** (AUC $\approx$ 0.32–0.34), close to EDR ($\approx$ 0.30–0.32): the geometry result is not an artifact of parcellation.

Second, at fsaverage5 the connectome modes bundled with Pang's demo are a synthetic surrogate (degenerate spectrum, near-zero reconstruction), so the only well-posed contrast *there* is geometry vs. EDR (shape vs. distance). It shows a small but significant shift of the *opposite* sign to the parcellated geometry-

Table 2: Within-subject shift in the geometric−connectome advantage (drug − control). Permutation is the exact sign-flip test.

Dataset	Data	ΔAUC	paired t (p)	Wilcoxon p	perm. p	d_z
LSD (n=12)	Raw	+0.0058	4.60 (0.0008)	0.001	0.001	1.33
	Denoised	+0.0044	3.5 (0.002)	0.002	0.003	1.19
Psilocybin (n=7)	Raw	+0.0103	2.30 (0.061)	0.031	0.031	0.87
	Denoised	+0.0061	1.68 (0.145)	0.078	0.063	0.63

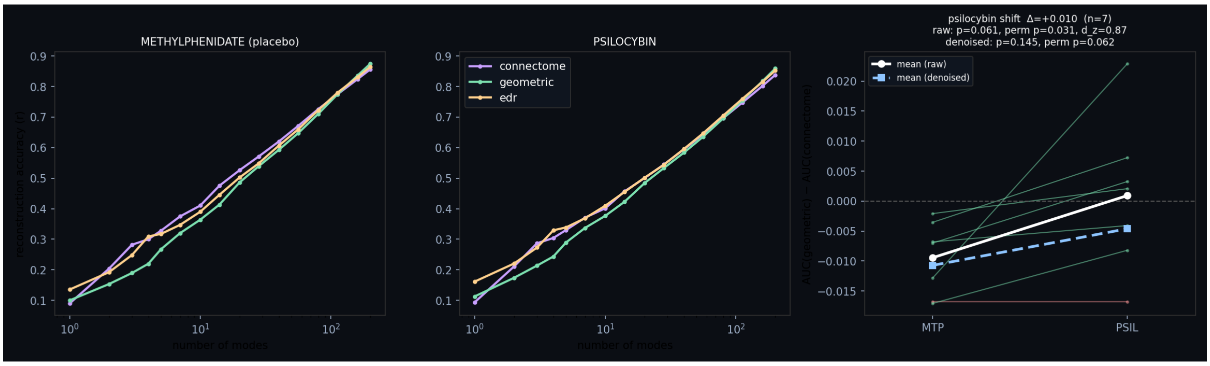

Figure 4: Replication (psilocybin, ds006072, $n = 7$). Same per-subject paired test on an independent drug, scanner, and pipeline: reconstruction accuracy under methylphenidate (left) and psilocybin (middle), and the per-subject shift (right). Activity again moves toward geometry under psilocybin (ΔAUC $= +0.010$, $d_z = 0.87$, exact-permutation $p = 0.031$).

vs-connectome shift: $\Delta(\text{geometric} - \text{EDR}) = -0.0067$, 95% CI $[-0.011, -0.002]$, paired $t(6) = -3.55$ ($p = 0.012$), Wilcoxon/permutation $p = 0.031$, $d_z = -1.34$. The "toward geometry" effect is thus *specific to the geometry-vs-wiring contrast* and does not generalize to geometry-vs-distance at fine scale.

Geometry versus the empirical connectome at vertex resolution. The synthetic-surrogate limitation can in fact be overcome: the Pang et al. [4] data release additionally provides the *empirical* high-resolution group-average tractography connectome (an $S = 255$ HCP average, 29,696 × 29,696 over the fsLR-32k left-hemisphere cortex). We compute its connectome-harmonic (normalized graph-Laplacian) eigenmodes directly and pit them against Pang's published fsLR-32k geometric eigenmodes, both evaluated on the same fsLR-32k grayordinates as the psilocybin CIFTI BOLD—so the data require no resampling. This yields a genuine geometry-versus-*wiring* comparison at vertex resolution (Figure 6). Two results follow. (i) The **geometric basis reconstructs cortical activity better than the real connectome in both states** (mean AUC: psilocybin geometric 0.428 vs. connectome 0.414; methylphenidate 0.470 vs. 0.460), confirming the geometric advantage of [4] off the parcellation and with real wiring, and removing the parcellated near-tie. (ii) The **drug-induced shift toward geometry is directionally consistent but not significant** at vertex resolution: per-subject AUC(geometric) − AUC(connectome), psilocybin minus methylphenidate, $\Delta = +0.0040$, 95% CI $[-0.0039, +0.0119]$, paired $t(6) = 1.23$ ($p = 0.26$), exact permutation $p = 0.20$, $d_z = 0.47$ (same direction after global-signal regression, $\Delta = +0.0037$, $p = 0.31$). With $n = 7$ the design is underpowered for an effect of this magnitude, so the psychedelic *shift* remains a parcellated/LSD-level result, whereas the baseline geometry-over-wiring advantage now holds at full vertex resolution with the empirical connectome.

3.6 Experiment 4: harmonic consonance decreases under psychedelics

A natural extension asks whether the degree of geometric/harmonic order relates to affect, as the Symmetry Theory of Valence [7] proposes. We cannot test this correlation directly — neither public release ships per-subject affect ratings — so we test the prerequisite *mechanism direction*: does a serotonergic psychedelic move the harmonic spectrum toward consonance/symmetry, or away from it? The answer is unambiguous and runs *against* the naive STV prediction (Figure 7). Under LSD the connectome-

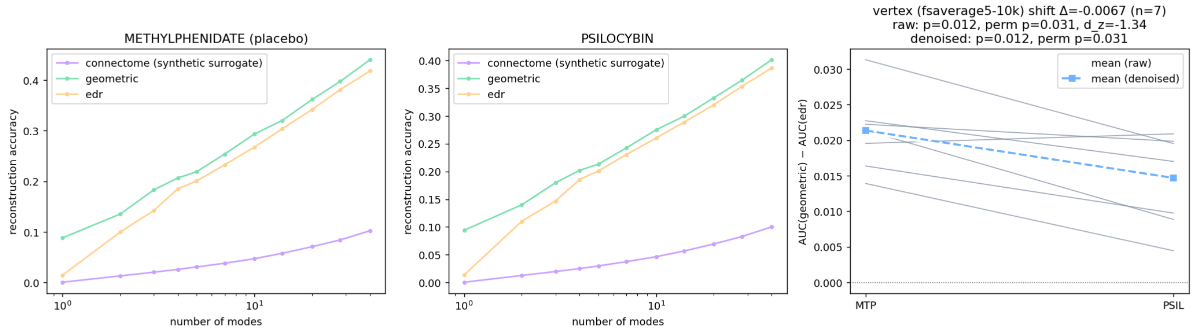

Figure 5: Vertex-resolution geometry vs. EDR (psilocybin, fsaverage5-10k, $n = 7$, left hemisphere). Reconstruction accuracy under methylphenidate (left) and psilocybin (middle) for the geometric and EDR bases (the public connectome basis at fsaverage5 is a synthetic surrogate that reconstructs near zero and is shown only for reference). Right: per-subject geometric−EDR advantage, which shifts slightly toward EDR under psilocybin ($\Delta = -0.0067$, permutation $p = 0.031$).

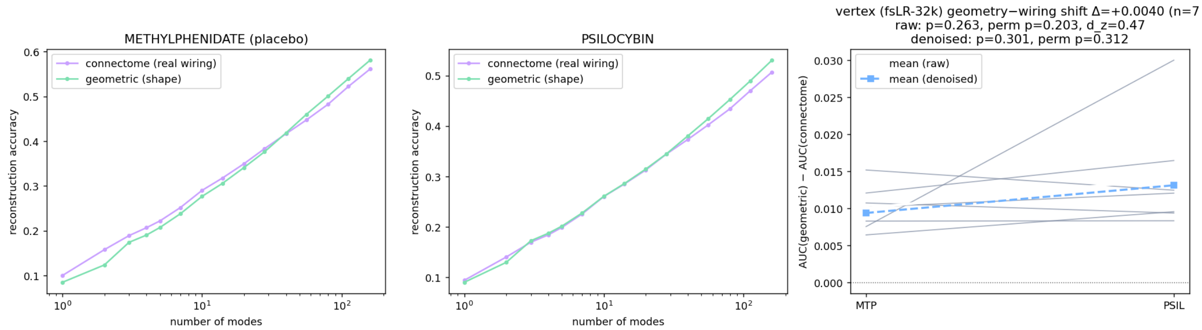

Figure 6: Geometry vs. the empirical connectome at vertex resolution (psilocybin, fsLR-32k, $n = 7$, left-hemisphere cortex). Reconstruction accuracy under methylphenidate (left) and psilocybin (middle) for the geometric basis and the connectome-harmonic basis computed from Pang's empirical $S255$ high-resolution group-average tractography connectome. The geometric (shape) basis reconstructs activity better than the real connectome in both states. Right: per-subject geometric−connectome advantage, methylphenidate vs. psilocybin; the shift toward geometry is directionally consistent but not significant at this sample size ($\Delta = +0.0040$, $d_z = 0.47$, permutation $p = 0.20$).

harmonic spectrum becomes markedly **more dissonant** (drug−placebo $\Delta = +16.7$, paired $p = 0.001$, exact permutation $p = 0.002$, $d_z = 1.24$), **broader** (spectral entropy $p = 0.007$), and shifted toward **higher** spatial frequencies (centroid $p = 0.015$); all three survive global-signal regression and frame censoring. Psilocybin ($n = 7$) trends in the same direction on every metric but does not reach significance (dissonance $\Delta = +11.0$, $d_z = 0.49$, $p = 0.24$). On the geometric-harmonic substrate the LSD dissonance increase is weaker but same-signed ($\Delta = +2.1$, $p = 0.022$).

This is the "entropic brain" direction [2] and clarifies a subtle point: the drift toward the smooth geometric *basis* in Experiment 3 is *not* the same as increased *consonance*. Activity can gain on the wiring basis relative to shape while the overall power spectrum simultaneously spreads to higher, less harmonically related modes. If anything, a simple spatial-consonance reading of STV predicts the wrong sign for the (typically positively-valenced) acute psychedelic state; we discuss the caveats to that inference below.

4 Discussion

Across two independent psychedelic datasets, cortical activity shifts toward a smoother, shape-aligned (geometric) description and away from one tied to specific structural wiring. The LSD effect is strong and survives our nuisance model; the psilocybin effect, in a smaller and harder (active-placebo) design, replicates the direction and effect size and is significant by exact nonparametric tests on raw data. Together with the spectral reorganization of Experiment 1 (less low-frequency, wiring-bound structure under LSD) and the unsupervised geometric latent space of Experiment 2, the picture is coherent:

psychedelics appear to relax activity off the connectome's specific scaffolding toward the broad geometric modes that the cortex's shape most readily supports.

Consonance and valence. Experiment 4 adds an important qualification. The same psychedelic state that aligns activity with smooth geometric modes also makes the harmonic power spectrum *less* consonant and broader — the opposite of what a literal spatial-consonance reading of the Symmetry Theory of Valence [7] would predict for a typically positive state. We stress that this is not a refutation of STV: we measure *spatial*-harmonic, not temporal, consonance; the acute psychedelic state is variably (not uniformly) valenced; and the "neural annealing" formulation of STV explicitly predicts a transient rise in energy and entropy ("heating") before any annealing toward order, so a snapshot of increased breadth during the drug condition is consistent with that account. Most importantly, without per-subject affect ratings — absent from both public datasets — we test only the mechanism direction, not the valence correlation itself. The result is best read as a sharp, falsifiable prediction generator: it shows that "more geometric alignment" and "more harmonic consonance" are dissociable, and that a proper STV test will require datasets pairing connectome-harmonic decompositions with validated affect scales.

Limitations. All results are group-average and/or parcellated at 400 regions — precisely the regime in which the geometric advantage over wiring is weakest, and in which geometric modes are hemisphere-separable while connectome and EDR modes are whole-brain. We therefore frame Experiment 3 as a test of the drug-induced *shift*, not as a resolution of the broader geometry-versus-wiring debate, which requires vertex-resolution surfaces and carefully constructed connectomes [5]. Our vertex-resolution analysis (Section 3.5), using Pang's empirical group-average connectome at fsLR-32k, confirms the geometry-over-wiring *advantage* at fine resolution, but with $n = 7$ is underpowered to confirm the drug-induced *shift*, which therefore stays a parcellated-level finding. The LSD data lack fmriprep motion parameters, so no full 24-parameter motion regression is possible; after denoising neither basis significantly beats the other within a single state (the claim is about the shift). Sample sizes are modest ($n = 12$ and $n = 7$), and the psilocybin cohort uses an active stimulant placebo, which makes its design conservative but underpowered. These are reanalyses of public data and are not clinical or individual-level findings, and not medical advice.

Conclusion. A simple, reproducible, parcellated pipeline shows that the relative fit of cortical *shape* versus *wiring* to brain activity is state-dependent and bends toward geometry under both LSD and psilocybin. We release all code, fetch scripts, precomputed results, and an interactive demo to support scrutiny and extension to vertex-resolution data.

Data and Code Availability

Code, analysis scripts, precomputed results, and an interactive web demonstration: `https://github.com/fractastical/connectome_harmonics`. Primary data are public: HCP-YA structural connectivity (Human Connectome Project terms apply), OpenNeuro `ds003059` (LSD), and OpenNeuro `ds006072` (psilocybin). Cortical surfaces, parcellations, and eigenmode tooling from NSBLab/BrainEigenmodes and ThomasYeoLab/CBIG retain their own licenses.

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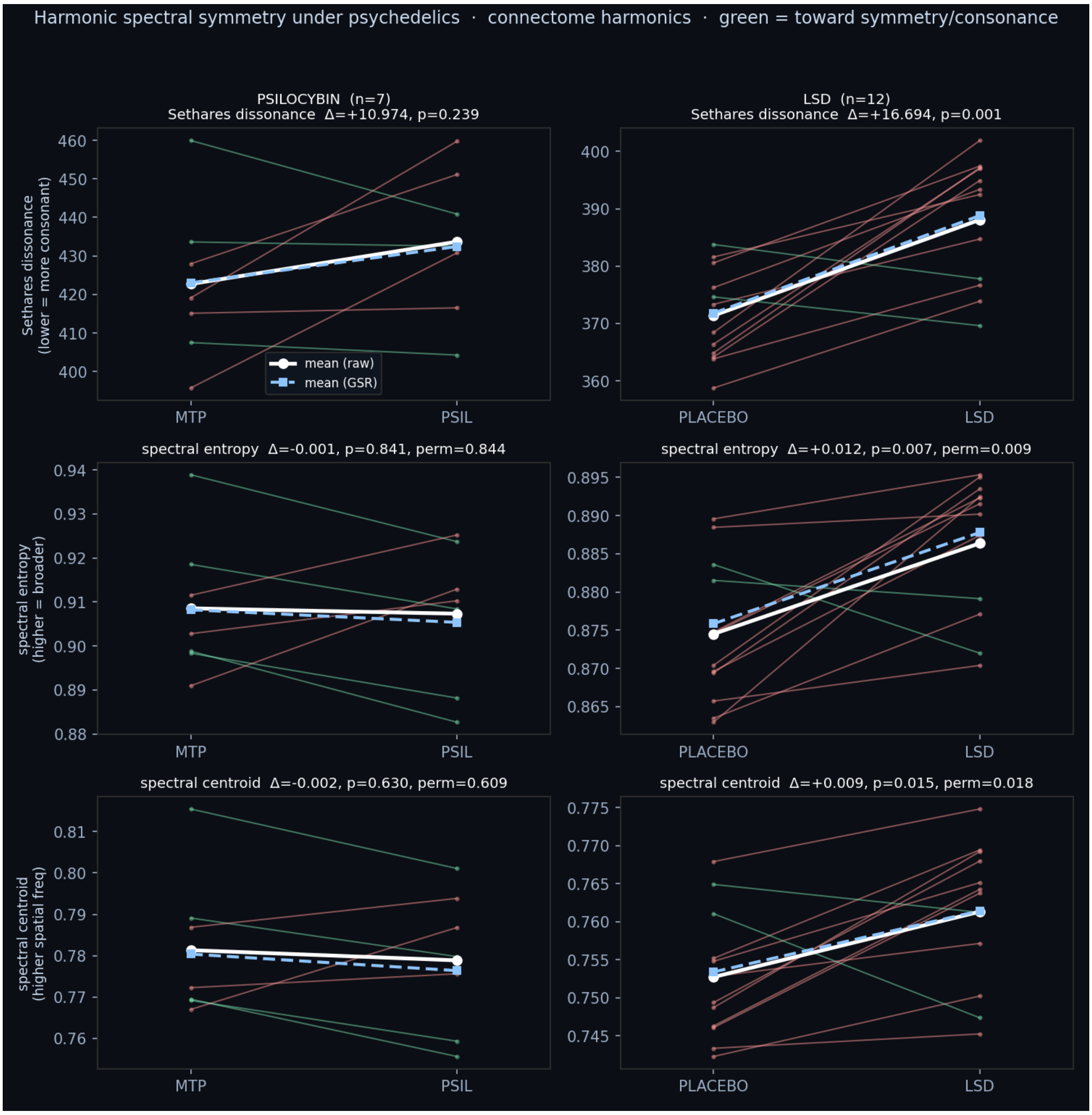

Figure 7: Harmonic consonance / spectral symmetry under psychedelics (connectome harmonics). Per-subject paired change (placebo→drug) in Sethares dissonance (top), spectral entropy (middle), and spectral centroid (bottom), for psilocybin (left, $n = 7$) and LSD (right, $n = 12$); green lines move toward symmetry/consonance, red away. White = raw means, blue dashed = after GSR. Under LSD nearly every subject moves away from consonance (more dissonant, broader, higher-frequency); psilocybin trends the same way.